

Predicting the Zero-Shot Transfer of a Promptable Video Tracker

Forecasting SAM 3's per-species, per-place reliability —
for detection and association — from label-free distances

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Abstract

Promptable trackers such as SAM 3 segment and track any named animal in a video with no training on that species, yet no rule tells a practitioner *in advance* whether the model is reliable for a given animal in a given place. This dissertation builds and validates a *predictive* rule for that reliability on the SA-FARI benchmark, whose splits are disjoint by species and location. We measure SAM 3’s zero-shot promptable-tracking accuracy with the official evaluator, decompose it into detection (**pDetA**) and association (**pAssA**), and test whether each can be predicted from four label-free distances (taxonomic, visual, environment, temporal). *[TODO: headline result + validated out-of-sample performance.]*

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Chapter 1

Introduction

1.1 Motivation

Promptable video trackers such as SAM 3 [4] segment and track any named animal in a video with no training on that species, and conservation teams now deploy them on new species and regions constantly. Yet no rule tells a practitioner, *before running the model*, whether its output is trustworthy for a given animal in a given place.

1.2 Problem statement and research question

This dissertation asks whether SAM 3’s zero-shot tracking accuracy on an unseen (species, place) can be *predicted in advance* from properties measurable without any label of the target, and which factors govern that transfer — separately for *finding* the animal and *following* it.

1.3 Hypotheses

H0 (null)

the label-free distances carry no predictive information about pDetA/pAssA — transfer is not distance-driven.

H1 (detection)

pDetA falls with species novelty (taxonomic + visual distance).

H2 (association)

pAssA falls with environment difficulty (scene, day/night/IR, clutter, camera motion).

Failing to reject H0 would itself be an informative result, motivating a pivot from distance-based prediction to representational probing (does SAM 3 encode taxonomy?).

1.4 Contributions

To our knowledge this is the first study to *predict*, before the model is run and from label-free distances, the zero-shot transfer of a promptable video tracker — and the first to split that prediction into **detection** (pDetA) and **association** (pAssA). This distinguishes it from label-free performance prediction for classifiers, which estimates a single scalar from the model’s own outputs on the target [1, 6, 10], and from the very recent static-image detector/segmenter estimators [14, 16]; and from camera-trap shift studies, which measure but do not forecast per-site reliability [2, 3, 7]. Concretely we deliver: (i) a validated, out-of-sample reliability estimator for SAM 3 over SA-FARI; (ii) the detection-versus-association decomposition, tested on two near-orthogonal splits — a constructed species hold-out (novelty) and the official location hold-out (environment); and (iii) an honest accounting of when distance *fails* to predict transfer (H0).

1.5 Dissertation outline

Chapter 2 reviews the related work; Chapter 3 describes the data, measurement, distance features, and modelling; Chapter 4 reports the results and out-of-sample validation; Chapter 5 discusses them; Chapter 6 concludes.

Chapter 2

Background and Related Work

This chapter places the dissertation among four bodies of work: promptable segmentation and tracking (the model we study), the tracking metric and its detection/association decomposition (how we score it), label-free prediction of out-of-distribution performance (the statistical idea we extend), and distribution shift in camera-trap vision (the application and the closest empirical neighbours). It closes by stating precisely the gap this work fills.

2.1 Promptable segmentation and tracking

The *Segment Anything* line moved segmentation from a fixed label set to an interactive, promptable formulation, and its successors extended prompting from points and boxes to *concepts*. SAM3 [4] is a promptable model that, given a short text prompt such as "impala", detects, segments, and tracks every matching instance across a video with no training on that category — a *zero-shot*, open-vocabulary detect–segment–track capability. It is exactly this capability that conservation teams now deploy on new species and regions, and it is the (frozen) model whose transfer we predict. SAM3 was released alongside SA-FARI [15], a large multi-animal camera-trap tracking dataset that we use as the substrate for our natural experiment (Section 2.6). We treat SAM3 strictly as a fixed black box: we never fine-tune it, and the only quantity we fit is a small regression *about* its behaviour.

2.2 The tracking metric and its decomposition

Higher Order Tracking Accuracy (HOTA) [9] evaluates multi-object tracking by combining, over a range of localisation thresholds, a *detection* term (**DetA**: were the right objects found?) and an *association* term (**AssA**: were identities kept consistent over time?). SA-FARI scores promptable tracking with a promptable variant, **pHOTA**, that decomposes identically into **pDetA** and **pAssA**. This decomposition is the intellectual core of the dissertation: rather than predict a single scalar, we ask whether *finding* and *following* an animal transfer for *different reasons* — the former with how novel the animal is, the latter with how difficult the scene is — and we therefore model the two components separately throughout.

2.3 Predictable out-of-distribution generalisation

A now-mature literature shows that a model’s performance under distribution shift can be anticipated *without* target labels. Miller et al. [10] observe that out-of-distribution (OOD) accuracy is strongly linearly correlated with in-distribution accuracy across many models and shifts, and Baek et al. [1] show that the *agreement* between a pair of models tracks OOD accuracy closely enough to estimate it from unlabelled data alone. A parallel strand estimates accuracy from a

model’s own statistics on the unlabelled target: average thresholded confidence [6] calibrates a confidence threshold on source data and reads off target accuracy as the mass above it, while Deng and Zheng [5] regress classifier accuracy directly on a *distribution-shift feature* — the Fréchet distance between train and test feature sets — which is the closest methodological cousin to our own “regress the score on a distance” approach.

Two properties characterise essentially all of this work, and both are limitations we depart from. First, it targets **image classifiers** with a single scalar accuracy. The idea has only recently reached structured outputs: Yoo et al. [16] estimate object-detection performance without ground truth from the spatial consistency and confidence reliability of pre-NMS boxes, and reverse classification accuracy [14] predicts per-image segmentation quality by training a reverse model against a labelled reference. These are the closest structured-output predecessors, but both operate on *static single images*, neither concerns identity over time, and none decomposes the estimate into detection versus association. Second, every one of these methods still *runs the model on the target* and inspects its predictions, confidences, or agreement — it estimates the score of an *already-executed* model. We instead forecast from properties knowable *before the model is ever run* on the target.

2.4 Distribution shift in camera-trap and wildlife vision

That camera-trap models degrade at new locations is well established. *Terra Incognita* [2] showed that recognisers excellent at trained camera sites fail sharply at unseen ones; WILDS [8] consolidated this (via iWildCam) as a canonical real-world shift; and PanAf-FGBG [3] demonstrated that behaviour-correlated *backgrounds* causally drive the drop, proposing a latent-space mitigation. These works are the empirical backdrop for our environment axis — but they share two features that mark the gap. They **measure or mitigate** degradation after the fact rather than *predicting* per-site or per-species reliability in advance, and they study **classifiers or behaviour recognisers**, not promptable trackers, scored by a single accuracy with no detection/association split. Most tellingly, a recent unified study of camera-trap recognition over time [7] explicitly names “predicting when zero-shot models will succeed at a new site” as an open question — and stops at naming it. This dissertation builds the predictor that question asks for.

2.5 What should predict transfer

Our candidate predictors are motivated by representation-learning results. Vision foundation models encode taxonomic structure across the tree of life [13], which motivates a *taxonomic* distance and, because appearance and phylogeny are correlated, a *visual* distance; we compute the latter from mask-cropped DINOv2 [11] embeddings, with CLIP [12] as a robustness variant. The camera-trap location results above [2, 3] motivate an *environment* distance from the scene background, and per-video timestamps motivate a *temporal* gap. Crucially, all four are computable without any label of the target animal, and all are measured against a frozen reference set, so no target information can leak into a predictor.

2.6 The SA-FARI benchmark

SA-FARI [15] is a large open multi-animal camera-trap tracking dataset (thousands of densely annotated videos across roughly 80 annotated species with a seven-level taxonomy, hundreds of locations on four continents, and per-video timestamps), released with SAM 3 and scored by the official VEval evaluator. It ships with abundant *hard negatives* — prompts for absent species that must return nothing — which we keep, as they drive the false-positive side of detection. Two properties of the shipped split shape our design and merit a precise statement. Inspection

of the released annotations shows the train/test split is disjoint by **location** (test camera sites are unseen) but that its *species are largely shared* across the split; a genuine “unseen species” axis is therefore not directly available from the official split. Because SAM3 is frozen and zero-shot on *all* of SA-FARI, however, the split reflects nothing the model learned — it is only our choice of a *reference distribution* against which distances are measured. This licenses a species-disjoint *reference/probe partition* of the same videos for the primary (species-novelty) experiment, complemented by the official location split for the environment experiment; the two are near-orthogonal probes of the same data, and their construction is detailed in Chapter 3.

2.7 Summary and gap

Predictable OOD generalisation is established for classifiers scored by a scalar [1, 5, 6, 10] and, very recently, for static detectors and segmenters [14, 16]; camera-trap vision has thoroughly documented, but not forecast, location-driven degradation [2, 3, 7, 8]. No prior work predicts, *before running the model* and from label-free distances, the zero-shot transfer of a **promptable video tracker**, and none separates that prediction into **detection** and **association**. Filling exactly that gap — a validated, out-of-sample reliability estimator for SAM3 over SA-FARI’s species-by-place-by-time structure, decomposed into pDetA and pAssA — is the contribution of this dissertation.

Chapter 3

Data and Methodology

3.1 Dataset, splits, and the analysis unit

We use SA-FARI [15] throughout. In the released `_ext` annotations, a probe is a (video, text-prompt) pair; the animal’s species and its seven-level taxonomy are resolved through the prompt’s `category_id`, while per-video `location_id` and creation timestamp give place and time. We aggregate scores to a *cell* — a (species, location, time) group — and attach support counts (n_{frames} , n_{masklets} , n_{videos}) used for weighting and as covariates. Hard negatives (prompts for absent species) are kept, as they drive the false-positive side of detection.

Two complementary splits. Inspection of the annotations shows the shipped split is disjoint by *location* but shares species across train and test (Section 2.6); on that split species-distance is ≈ 0 for every cell, so it cannot test species novelty. Because SAM3 is frozen and zero-shot on all of SA-FARI, the split only defines our *reference distribution*, so we are free to repartition the same videos. We therefore use two near-orthogonal experiments and run inference only once (scoring is per-video and split-agnostic):

Split A — species hold-out (primary).

The pooled species are partitioned into a reference set and a held-out set, stratified by taxonomic group so held-out species span a range of novelty. Species-distance varies while the environment stays largely familiar (held-out species recur at reference locations), isolating H1. Distances use a *leave-species-out* reference and it is validated by leave-species-out cross-validation.

Split B — location hold-out (secondary).

The official train/test split, whose unseen locations isolate H2 and whose benchmark comparability carries the Phase-1 score-sanity check.

Split A is a *constructed* hold-out — legitimate because the model is frozen, and reported as such; we cross-check aggregate scores against Split B, which is benchmark-comparable.

3.2 Measuring transfer

SAM3 [4] is run frozen over the probe videos (species-specific prompts as the primary condition, a generic “animal” prompt for robustness) and scored with the official `VEval` evaluator — the metric is never re-implemented — giving per-cell `pDetA`/`pAssA`/`pHOTA` [9] with their support. A single inference pass over the union of Split A and Split B probes serves both experiments.

3.3 The frozen reference (leakage firewall)

For each split the reference (“seen”) species, locations, and taxonomy are frozen once; every distance is then computed against that reference only, so no probe information can enter a

feature — verified by a unit test that swaps the reference and checks the distances move as expected. We assert the axis that is genuinely disjoint (location for Split B; the held-out species for Split A) and *report* any residual species overlap honestly rather than asserting it away. On Split A, distances use a leave-species-out view of the reference, so a held-out species’ distance is to the nearest *other* reference species.

3.4 Label-free distance features

All four distances (plus a proxy) are label-free for the target and computed against the frozen reference.

3.4.1 Taxonomic distance

Lowest-common-ancestor tree steps from the probe’s species to the nearest reference species (shared genus = 1, family = 2, order = 3, ...); the leave-species-out reference keeps this non-trivial for shared species.

3.4.2 Visual distance

Cosine distance between DINOv2 [11] embeddings of the *ground-truth-mask-cropped* animal and the nearest reference-species prototype, with a Fréchet/MMD variant and a CLIP [12] encoder as robustness checks. Using the dataset’s ground-truth masks means this feature needs no SAM 3.

3.4.3 Environment distance

Embedding distance between the animal-masked-out background scene and the nearest reference location, plus an interpretable day/night–infrared flag derived from colour statistics.

3.4.4 Temporal gap

The minimum absolute year gap between a cell’s footage and the nearest reference footage.

3.4.5 SAM 3 familiarity proxy

Separability of the target species in SAM 3’s own feature space — a hedge against the unknown true pretraining set, reported alongside the other distances rather than treated as ground truth.

3.5 Modelling the law

Each bounded target (pDetA, pAssA) is modelled *separately* with a support-weighted beta / logit-link regression that includes a $\log(n_{\text{frames}})$ covariate, so that “rare” is never mistaken for “far”. Variance is attributed to each factor by dominance/commonality analysis with a VIF collinearity check — not by four univariate fits — and the headline contrast (does detection load on species novelty and association on environment?) is quantified across the two targets. The fitted models are validated *out of sample* by holding out whole groups: leave-species-out on Split A and leave-location-out on Split B, with bootstrap confidence intervals reported wherever a point estimate appears. The validated models are wrapped into the reliability estimator: four distances in, predicted pDetA/pAssA with an interval out.

3.6 Reproducibility and constraints

The model is frozen and scored by the official evaluator; the seen set is frozen before any distance is computed; hard negatives are retained; and all inference is single-GPU and inference-only. Every reported number is regenerated from the repository, and the split definitions, seeds, and reference sets are persisted so both experiments are reproducible end to end.

Chapter 4

Results and Evaluation

4.1 Measurement

4.2 The fitted law

4.3 Variance partitioning and the decomposition test

4.4 Out-of-sample validation

4.5 Ablations and robustness

Chapter 5

Discussion

5.1 Interpretation

5.2 A deployment reliability estimator

5.3 Limitations and threats to validity

5.4 Ethical considerations

Chapter 6

Conclusion

6.1 Summary of findings

6.2 Contributions

6.3 Future work

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Appendix A

Reproducibility

Appendix B

Supplementary Results